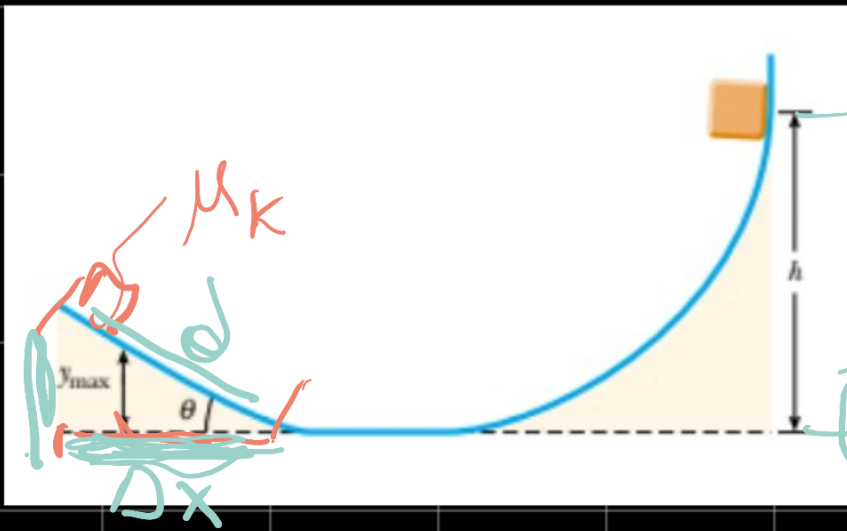




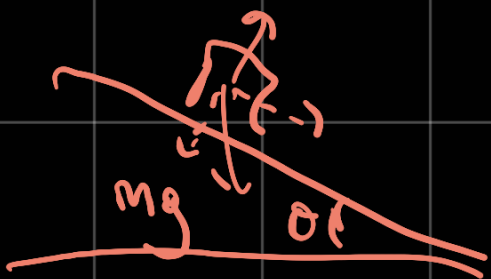
$$v_0 = 0$$

$$y_{\max} = \frac{h}{1 + \mu_k \cot \theta}$$

$$(y = 0)$$

$$\Rightarrow \Delta E = W \Rightarrow \Delta U = W$$

$$\Rightarrow mgy_{\max} - mgh = -\underline{f_k} \cdot d$$



$$\Rightarrow mgy_{\max} - mgh = -\mu_k N \cdot d$$



$$\sin \theta = \frac{y_{\max}}{d}$$

$$\Rightarrow d = \frac{y_{\max}}{\sin \theta}$$

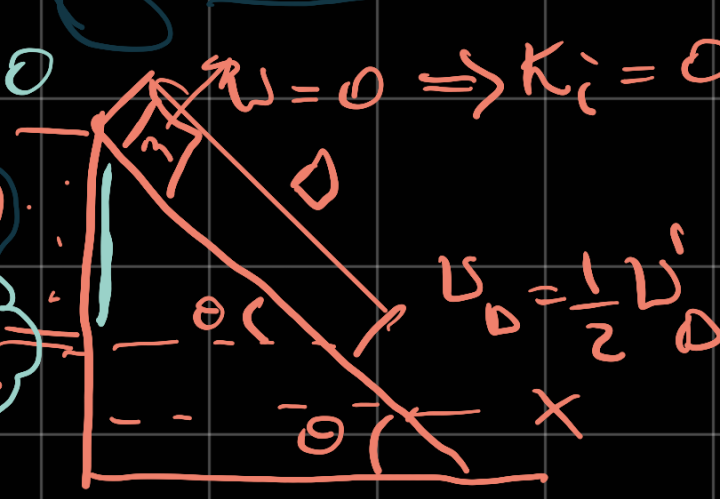
$$\Rightarrow mgy_{\max} - mgh = -\mu_k mg \cos \theta \cdot d$$

$$\Rightarrow y_{\max} - h = -\mu_k y_{\max} \cot \theta$$

$$\Rightarrow y_{\max} (1 + \mu_k \cot \theta) = h$$

$$\Rightarrow y_{\max} = \frac{h}{1 + \mu_k \cot \theta}$$

$y=0$
 $\mu_k = \frac{3}{4} \tan \theta$
 $v_0 = \frac{1}{2} v'_0$
 $y=0 \Rightarrow$



Fricción: $\Delta E = -f_k \cdot D \rightarrow W$

$$\Rightarrow \Delta U + \Delta K = -f_k \cdot D \quad \Delta U = U_f - U_i$$

$$\Rightarrow -mgD \sin \theta + \frac{1}{2} m v_D^2 = -\mu_k mg \cos \theta D$$

$$\Delta E = 0 \Rightarrow K_f + U_f = K_i + U_i$$

$$\Rightarrow \frac{1}{2} m v_0^2 = mgD \sin \theta - \mu_k mg \cos \theta D$$

$$\Rightarrow v_D^2 = 2gD (\sin \theta - \mu_k \cos \theta) \quad *$$

Sin Fricción: $\Delta E = 0$
 $\rightarrow E_f = E_i$

$$\Rightarrow K_f = U_{gi} \Rightarrow \frac{1}{2} m v_D'^2 = m g D \sin \theta$$

$$\underline{v_D'^2 = 2 g D \sin \theta} \quad (*)$$

$$v_D = \frac{1}{2} v_D' \quad \swarrow$$

$$\Rightarrow \left(\frac{v_D'}{2} \right)^2 = 2 g D (\sin \theta - \mu_k \cos \theta)$$

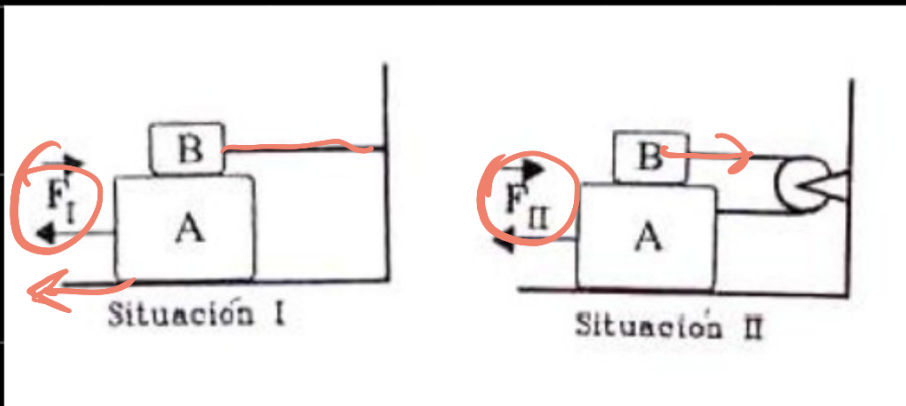
$$\Rightarrow \frac{v_D'^2}{4} = 2 g D (\sin \theta - \mu_k \cos \theta)$$

$$\Rightarrow \frac{g \cancel{D} \sin \theta}{2} = 2 \cancel{g D} (\sin \theta - \mu_k \cos \theta)$$

$$\Rightarrow \frac{1}{2} \sin \theta = 2 (\sin \theta - \mu_k \cos \theta)$$

$$\Rightarrow -\frac{3}{2} \sin \theta = -2 \mu_k \cos \theta$$

$$\Rightarrow \mu_k = \frac{3}{4} \tan \theta$$

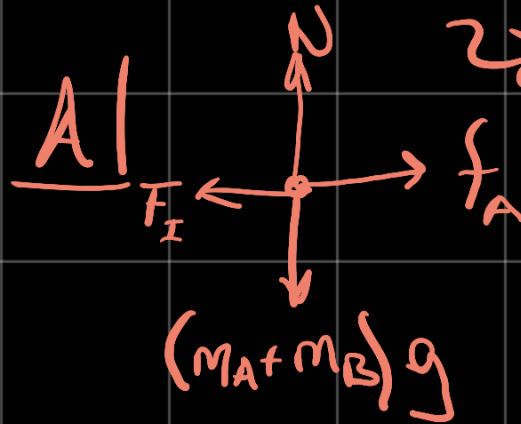


$$\begin{aligned} \underline{F_I} &\rightarrow ? \text{ I} \\ \underline{F_{II}} &\rightarrow \text{ II} \end{aligned}$$

$$N = \text{cte} \Rightarrow a = 0$$

$$m_A = 2m_B$$

Situación I:



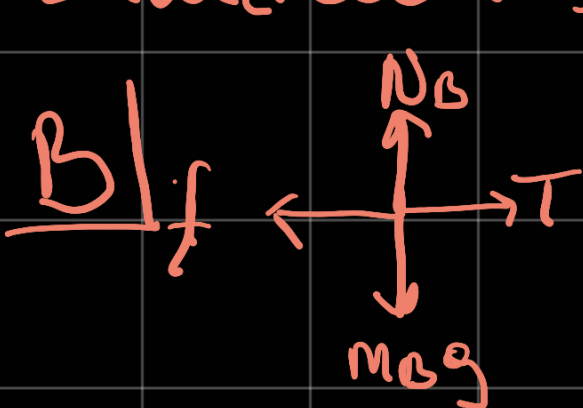
$$\begin{aligned} \sum \vec{F} &= 0 \quad \sum F_y = 0 \Rightarrow N - (m_A + m_B)g = 0 \\ N &= (m_A + m_B)g \end{aligned}$$

$$\sum F_x = 0 \Rightarrow F_I - f_A = 0$$

$$\Rightarrow F_I = f_A = \mu N = \mu (m_A + m_B)g$$

$$F_I = \mu (2m_B + m_B)g = 3\mu m_B g$$

Situación II



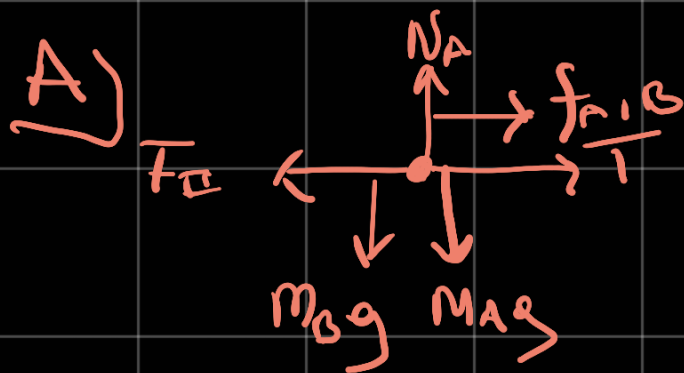
$$\sum \vec{F} = 0$$

$$\sum F_x = 0$$

$$T - f = 0 \Rightarrow T = f$$

$$\sum F_y = 0 \Rightarrow N_B - m_B g = 0$$

$$\Rightarrow T = \mu N = \mu m_B g$$



$$\sum \vec{F} = 0$$

$$\sum F_x = 0$$

$$f_{A,B} = \mu(m_A + m_B)g$$

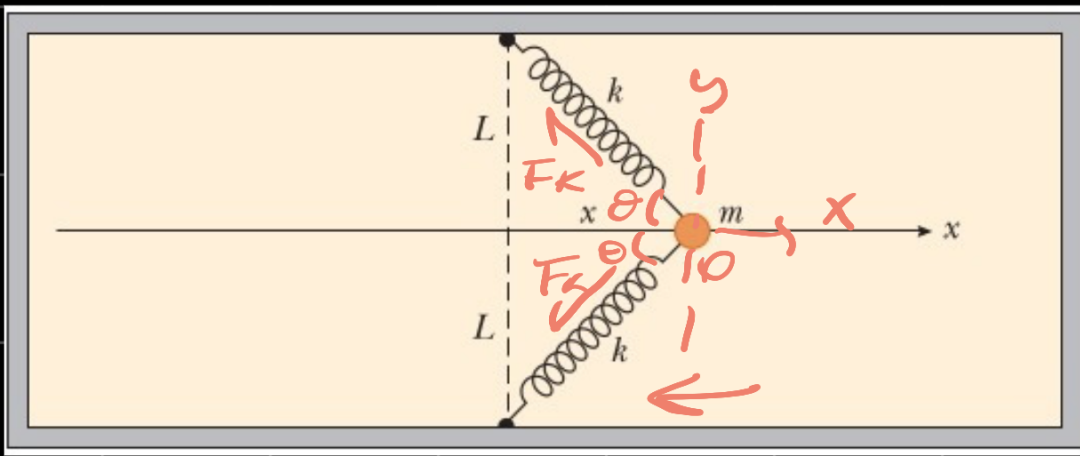
$$\Rightarrow T - F_{II} + \mu(m_A + m_B)g = 0 \quad m_A = 2m_B$$

$$\Rightarrow F_{II} = T + \underbrace{\mu(m_A + m_B)g}_{3m_B g}$$

$$= \mu m_B g + 3\mu m_B g$$

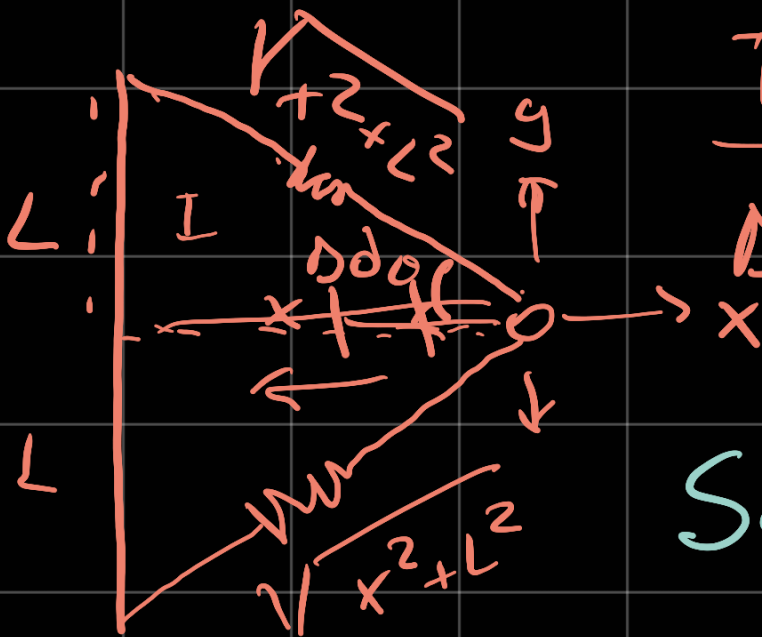
$$\Rightarrow F_{II} = 4\mu m_B g$$

$$\Rightarrow \frac{F_{II}}{F_{II}} = \frac{3\mu m_B g}{4\mu m_B g} = \frac{3}{4}$$



No se
evalúa

$$a) \vec{F} = -2kx \left(1 - \frac{L}{\sqrt{x^2 + L^2}} \right) \hat{i}$$



$$F_k = k \Delta l$$

$$\Delta l = \sqrt{x^2 + L^2} - L$$

Simetría

$$\sum \vec{F} \quad \sum F_x = -F_k \cos \theta - F_k \cos \theta = -2F_k \cos \theta$$

$$\sum F_y = 0 \Rightarrow F_k \sin \theta - F_k \sin \theta = 0$$

$$\Rightarrow \cos \theta = \frac{\Delta l}{\text{Hip}} = \frac{x}{\sqrt{x^2 + L^2}}$$

$$\Rightarrow F = F_x = -2F_k \cos \theta$$

$$= -2K(\sqrt{x^2 + L^2} - L) \frac{x}{\sqrt{x^2 + L^2}}$$

$$\Rightarrow \vec{F} = -2Kx \left(1 - \frac{L}{\sqrt{x^2 + L^2}}\right) \hat{L}$$

$$b) U(x) = Kx^2 + 2KL(L - \sqrt{x^2 + L^2})$$

Función de Energía potencial

$$\Delta U = - \int_{x_i}^{x_f} \vec{F}_x dx$$

$$U(x) = + \int_0^x 2Kx \left(1 - \frac{L}{\sqrt{x^2 + L^2}}\right) dx$$

$$= 2K \int_0^x \left(x - \frac{Lx}{\sqrt{x^2 + L^2}}\right) dx$$

$$= 2k \left\{ \int_0^x x \, dx - L \int_0^x \frac{x}{\sqrt{x^2 + L^2}} \, dx \right\}$$

I_1 I_2

formula

$$\underline{I_1} \quad \int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$I_1 = \int_0^x x' \, dx = \frac{x^2}{2} \Big|_0^x = \frac{x^2}{2} - \frac{(0)^2}{2}$$

$$\underline{I_2} \quad \int_0^x \frac{x}{\sqrt{x^2 + L^2}} \, dx \quad \left(\begin{array}{l} u = \sqrt{x^2 + L^2} \\ du = \frac{1}{2}(x^2 + L^2)^{-1/2} \cdot 2x \, dx \\ = \frac{x}{\sqrt{x^2 + L^2}} \, dx \end{array} \right)$$

$$\Rightarrow \underline{dx} = \frac{\sqrt{x^2 + L^2}}{x} \, du$$

$$u(0) = \sqrt{(0)^2 + L^2} = L \quad \left. \vphantom{\int_0^0} \right\} \int_0^0$$

$$u(x) = \sqrt{x^2 + L^2}$$

$$\Rightarrow \int_L^x \frac{\sqrt{x^2+L^2}}{\sqrt{x^2+L^2}} du$$

$$\Rightarrow \int_L^x du = u \Big|_L^x$$

$$\Rightarrow \sqrt{x^2+L^2} - L$$

$$U(x) = 2K \left\{ \frac{x^2}{2} - L(\sqrt{x^2+L^2} - L) \right\}$$

$$= Kx^2 - 2KL(\sqrt{x^2+L^2} - L)$$

$$\Delta E = W \rightarrow \text{system ref.}$$

$$\Rightarrow \cancel{\Delta K} + \Delta U_g + \Delta U_k = W$$

$$\Delta W_g = -\Delta U_g$$

$$-\Delta W_g = \Delta U_g$$

$$\mu_K \cdot 0 = 0$$